PROJECT BRIEF: BATTERY PCB DEVELOPMENT FOR BIONIC ARM

1. OBJECTIVE

Develop a custom PCB for a compact battery module, covering full electronic hardware design, firmware development (if required), and prototype embedding — for integration into a bionic arm socket.

2. SCOPE OF WORK

• PCB Design and Fabrication:

Compact, efficient PCB layout.

Integration of center ring RGB LED strip control.

USB Type-C charging input with fast charging capability.

Pass-through charging feature for external devices (e.g., mobile phones).

• LED Status Indication:

Smooth animated RGB lighting for battery charge levels:

Green (100–80%)

Yellow (79–30%)

Red (<30%).

• Sound Alert (Optional):

- Beep or buzzer-based alerts for low battery and full charge status (optional).

• Battery Management System (BMS):

- Protection circuits for overcharge, over-discharge, and short-circuit events.

• Feature Highlight:

The USB Type-C port supports reverse power output, enabling users to conveniently charge their mobile phones directly from the prosthetic arm socket."

3. DELIVERABLES

- Complete PCB schematics and Gerber files.
- Bill of Materials (BOM) with full part specifications.
- Firmware (if required) for LED control and sound alerts.
- Prototype samples for functional testing.

Note:
Vendor is not responsible for casing, industrial design, or final product assembly!

Please let us know if you require any additional information, specifications, or support from our side to proceed with the development smoothly.